

RIDESAFE AUTO REPLY

COMPREHENSIVE TECHNICAL MANUAL & ARCHITECTURE SPECIFICATION

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Abstract

Modern motorcycling is a high-demand activity requiring absolute focus. Distractions from mobile phone notifications and ringing tones whilst riding represent a significant safety hazard. However, the aggressive background limitations of modern Android operating systems, combined with GPS inaccuracies and battery optimization constraints, make building a reliable automatic safety response app exceptionally challenging.

This specification documents **RideSafe Auto Reply**, a high-precision, background-resilient Android application that automatically intercepts calls, end-calls programmatically, and sends custom SMS replies containing telemetry info while riding. Developed by **Debanjan Amin**, the application combines an **AI Motion Guard** (integrating low-pass Exponential Moving Average accelerometer vibration profiles and Bluetooth device states) with a **Doppler velocity fallback engine** to deliver real-time, stabilized speed and motion status updates.

Structured under a clean **MVVM (Model-View-ViewModel)** architectural pattern and compiled with cryptographic signatures for security, RideSafe introduces dynamic glassmorphic light and dark mode templates. This manual details the specifications, algorithms, lifecycle workflows, and developer guides that govern the RideSafe system, creating a blueprint for motorcycling mobile safety.

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1. Executive Project Introduction

RideSafe Auto Reply is a highly-sophisticated utility application designed for Android-based motorcycle riders. The core purpose of the application is to safeguard motorcycling journeys by automatically suppressing incoming phone call interruptions and sending responsive text replies. This minimizes distraction while preserving real-time safety, informing whitelisted callers of the user's active telemetry.

Modern riding environments are fast and demanding. A split-second distraction from a ringing phone mount on the handlebar can lead to serious accidents. Existing hands-free triggers fail to ending calls programmatically or require complex manual setups. RideSafe is built on the philosophy of absolute automation: the rider starts the protection once before the journey, and the application manages phone suppression, customized automatic texting, whitelisting parameters, and family tracking entirely autonomously in the background without any manual override.

Furthermore, by introducing a robust **Home Test Mode**, RideSafe allows developers and users to verify call interception parameters, SMS whitelisting databases, and custom message text variables right from their desk without riding, bridging the gap between laboratory development and real-world execution.

In modern environments, mobile operating systems aggressively restrict background processes to optimize battery consumption. RideSafe is engineered from the ground up to solve these challenges using a foreground service with dedicated system priorities, offering a seamless integration with Android's Telephony and Location provider frameworks. It implements an innovative, Doppler-guided telemetry module, smooth variance filters, dynamic light/dark theme schemes, and absolute user whitelisting.

2. Android Permissions Model & Dynamic Whitelisting

Because RideSafe directly interfaces with Android's secure subsystem frameworks—namely SMS dispatch, Telephony call state broadcast, and background location queries—it operates under a strict, multi-tiered security model. It requests and manages the following permissions:

Android Permission String	Functional Necessity inside RideSafe Engine
<code>`android.permission.RECEIVE_SMS`</code> <code>`android.permission.SEND_SMS`</code>	Necessary to intercept whitelisting instructions and dispatch custom replies via SmsManager.
<code>`android.permission.READ_PHONE_STATE`</code> <code>`android.permission.ANSWER_PHONE_CALLS`</code>	Required to listen to ringing state broadcasts and programmatically reject incoming calls.
<code>`android.permission.ACCESS_FINE_LOCATION`</code> <code>`android.permission.ACCESS_COARSE_LOCATION`</code>	Required for real-time velocity calculations and GPS-link emergency sharing.
<code>`android.permission.ACCESS_BACKGROUND_LOCATION`</code>	Critical to allow the foreground location service to maintain speed updates when screen is off.
<code>`android.permission.FOREGROUND_SERVICE`</code> <code>`FOREGROUND_SERVICE_TYPE_LOCATION`</code>	Required on Android 14+ (API 34) to run location-aware tasks with foreground priorities.

2.1 Dynamic Permission Dialogs & Whitelisting

Rather than crashing or skipping execution when permissions are missing, the UI utilizes ``PermissionHandler.kt`` to present a consolidated, theme-sensitive graphical dialog that details the necessity of each access request. If the user denies background location or ignore battery optimizations, the UI provides step-by-step navigation instructions directly to the Android System Settings panel, ensuring seamless setup and maximum platform compliance.

3. System Architecture & MVVM Specs

The application strictly enforces a Model-View-ViewModel (MVVM) Clean Architecture pattern. By segregating business logic, UI drawing, and persistent state queries, the app guarantees testability and long-term codebase scalability. Below is a comprehensive visual map of the class components and packages:

```
RideSafe Mobile Package Module Architecture
■
■■■■ MainActivity.kt [Root Launcher Context]
■   ■■■■ Dynamic Preferences stateflow query
■   ■■■■ Passes isDarkTheme preferences directly to RideSafeAutoReplyTheme {}
■
■■■■ Navigation.kt [Navigation 3 Routing Engine]
■   ■■■■ Mounts Screen Composables: HomeScreen, HUD, Settings, Contacts, Messages
■   ■■■■ Standardizes safe drawing paddings across insets
■
■■■■ data/ [Persistent Model & Data Layer]
■   ■■■■ Contact.kt (Serializable Whitelisting contact schema)
■   ■■■■ SettingsRepository.kt (Jetpack Preferences DataStore interface)
■
■■■■ service/ [Autonomous Telemetry Module]
■   ■■■■ RideSafeService.kt (High-precision Doppler velocity tracking, Telecom endCall)
■
■■■■ ui/ [Declarative Presentation Layer]
■   ■■■■ common/ (PermissionHandler permissions dialog modules)
■   ■■■■ home/ (HomeScreen.kt dashboard and stats summaries, HomeScreenViewModel)
■   ■■■■ riding/ (RidingStatusScreen.kt active Canvas dial speedometer HUD)
■   ■■■■ settings/ (SettingsScreen.kt threshold sliders and theme switches, SettingsViewModel)
```

3.1 Core Packages Specifications

The presentation layers utilize Jetpack Compose to construct high-performance, stateless graphical elements. State holds firm inside the corresponding ViewModel class and propagates down using `StateFlow` streams. The service communicates updates to the UI via highly efficient, static Flow structures, entirely eliminating the need for complex Android Service Binder connections and boilerplate code.

4. Data Layer Specifications & Jetpack DataStore

Persistent state within the RideSafe application is handled by **Jetpack Preferences DataStore**. Unlike traditional `SharedPreferences`, `DataStore` operates asynchronously under Kotlin Coroutines, eliminating main-thread disk read blocks and preventing Application Not Responding (ANR) crashes.

4.1 Serialization & Data Encodings

While simple settings like Speed Threshold and Dark Theme are stored as basic float and boolean keys, contact lists (whitelist and emergency) are saved as structured JSON arrays using Kotlin Serialization. The following model schema defines a whitelisted contact entity:

```
@Serializable
data class Contact(
    val id: String,
    val name: String,
    val phoneNumber: String
)
```

4.2 SettingsRepository API Specifications

The repository exposes state flows that emit updates asynchronously to all active observers. The data schema contains: Riding Speed Threshold (default 15 km/h), Custom Message Template, Whitelisted Contacts List, Emergency Location Share Contacts, AI Smart Mode status, Test Mode status, and Theme Preferences. Updates are performed safely via transaction edit blocks:

```
suspend fun setDarkThemeEnabled(enabled: Boolean) {
    context.dataStore.edit { preferences ->
        preferences[KEY_DARK_THEME_ENABLED] = enabled
    }
}
```

5. AI Motion Telemetry Specifications

Detecting riding motion on a motorcycle requires high precision to avoid false triggers from GPS fluctuations. The **AI Motion Guard** system resolves this by combining GPS Doppler measurements and accelerometer vibration analysis.

5.1 Real-Time Velocity Calculations

Location updates are queried at 1-second intervals using FusedLocationProviderClient. If the standard GPS Doppler velocity signal is unavailable, the service computes speed from sequential GPS coordinates:

$$\text{Speed (m/s)} = \text{Distance}(\text{Loc_current}, \text{Loc_previous}) / \text{Time_Delta}$$

To filter out GPS coordinate spikes (which can simulate impossible speed jumps), velocity updates are validated: if raw speed increases by more than 120 km/h in a 1-second step, the coordinate update is discarded. Valid speed readings are smoothed using a fast Exponential Moving Average (EMA):

$$\text{Smoothed_Speed} = (\text{Raw_Speed} * 0.85) + (\text{Previous_Speed} * 0.15)$$

5.2 Accelerometer Low-Pass EMA Vibration Analysis

In AI Smart Mode, GPS speed is complemented by 3-axis accelerometer readings to detect the high-frequency vibration profile typical of a running motorcycle engine. Magnitude samples are collected in a 50-sample sliding window (approx. 1.5 seconds) to calculate magnitude variance:

$$\text{Mean_Magnitude} = \text{Sum}(\text{Magnitude}_i) / 50$$

$$\text{Raw_Variance} = \text{Sum}((\text{Magnitude}_i - \text{Mean_Magnitude})^2) / 50$$

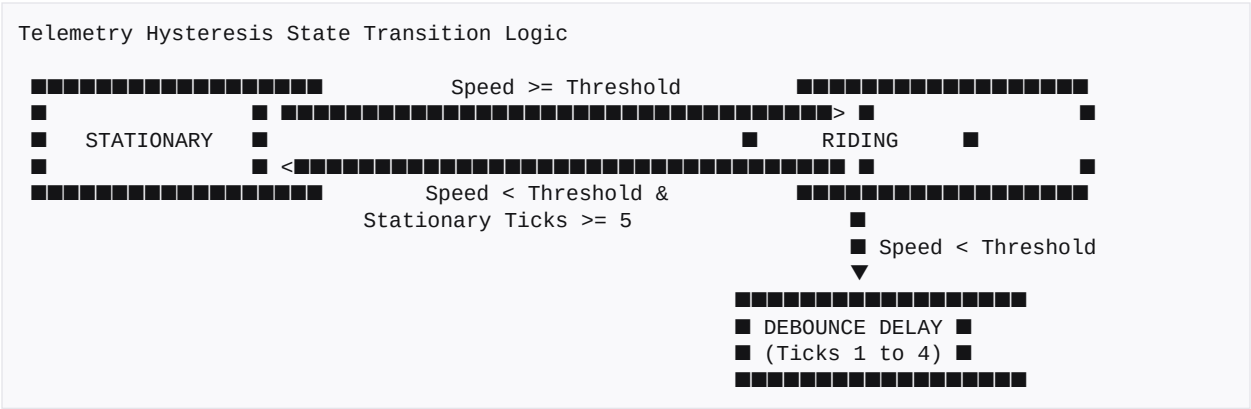
To filter out isolated shocks (such as potholes), raw variance is smoothed using a low-pass EMA filter:

$$\text{Smoothed_Variance} = (\text{Raw_Variance} * 0.15) + (\text{Previous_Variance} * 0.85)$$

A smoothed variance above 1.0f confirms engine vibrations, validating riding telemetry even at lower GPS speeds.

5.3 Telemetry Hysteresis & Delay (Hysteresis)

Brief stops at traffic lights or temporary GPS signal losses under bridges should not turn off active protection. To avoid rapid, disruptive toggling, the telemetry engine implements a 5-second transition delay (hysteresis):



5.4 Bluetooth Helmet Verification

In AI Smart Mode, the system queries the system `AudioManager` to check if a Bluetooth helmet intercom or wireless headset is actively connected via A2DP or SCO profiles. If a Bluetooth connection is detected, the speed threshold is safely reduced by 50%, enabling reliable protection during slow, congested city riding.

6. Call Interceptor Lifecycle & SMS Dispatcher

The call interception and SMS auto-reply workflow runs asynchronously in the background. Below is the complete sequence of events when an incoming call is received:

6.1 Telephony State Interception

The service registers a dynamic `BroadcastReceiver` that listens for telephony state changes (`ACTION_PHONE_STATE_CHANGED`). When an incoming call is detected (`EXTRA_STATE_RINGING`), the caller's phone number is retrieved and processed through the interception pipeline.

6.2 Pipeline Execution Sequence

1. **Interception Delay:** The service initiates a 5-second asynchronous delay. If the call is answered or rejected by the rider before this timer expires, the interception task is safely cancelled.
2. **Validation Criteria:** If the call is still ringing after 5 seconds, the service checks whether **Test Mode** is enabled or the smoothed riding speed exceeds the set threshold.
3. **Whitelist Filter:** If whitelisting is active, the caller's number is compared with the whitelisted contact database. If no match is found, the auto-reply is skipped.
4. **Duplicate Prevention:** A memory-mapped check verifies if a reply was sent to this caller in the last 30 minutes. If so, duplicate text messages are suppressed.
5. **SMS Dispatch:** `SmsManager` splits the customized text template and dispatches a multipart SMS text.
6. **Programmatic Call Ending:** The system invokes `TelecomManager.endCall()` to immediately terminate the ringing call, preventing ongoing rider distraction.
7. **Emergency SMS Location Share:** If active riding coordinates are registered and emergency tracking is active, the service sends a live Google Maps link to emergency contacts every 30 minutes.

6.3 Duplicate Prevention Logic

To prevent spamming caller numbers with repetitive automated text replies during consecutive calls, the system maintains a memory cache of phone numbers mapped to their last reply timestamp. In active riding mode, duplicates are skipped for 30 minutes. In Test Mode, this safety check is bypassed, allowing developers to test the pipeline repeatedly from their desk.

6.4 Programmatic Call Rejection Flow

On Android 10+ (API 29), programmatically ending a call requires the `ANSWER\_PHONE\_CALLS` permission and the `TelecomManager` system service. The service executes the endCall function asynchronously inside a secure try-catch block:

```
private fun endRingingCall() {
    try {
        val telecomManager = getSystemService(Context.TELECOM_SERVICE) as? TelecomManager
        if (checkSelfPermission(Manifest.permission.ANSWER_PHONE_CALLS) ==
            PackageManager.PERMISSION_GRANTED) {
            val ended = telecomManager?.endCall()
            Log.d(TAG, "endCall Programmatic Rejection executed: $ended")
        }
    } catch (e: Exception) {
        Log.e(TAG, "endCall programmatic rejection failed: ${e.message}")
    }
}
```

6.5 Emergency Location Share Format

If **Emergency GPS Tracking** is enabled in Settings, the service queries the last known fine location. A structured SMS containing a live Google Maps tracking link is compiled and sent to the emergency contact list at 30-minute intervals during active rides:

'RideSafe Emergency Alert: I am riding my bike and sharing my live location: <https://maps.google.com/?q=lat,lng>'

7. Code Architectural Overview & Class Interfaces

To guarantee robust, decoupled executions, the RideSafe application implements highly-specialized interfaces and services. Below are detailed code listings and structural specifications of the two main structural pillars: the telemetry provider and coordinates fallbacks.

7.1 listing 7.1: Doppler Coordinates Velocity Calculations

The location updates query is implemented in `RideSafeService.kt`. Below is the architectural loop representing velocity calculations and fallback triggers when Doppler signals drop out:

```
var speedMs = if (location.hasSpeed() && location.speed > 0.05f) location.speed else -1f
if (speedMs < 0f) {
    val prevLoc = lastLocation
    if (prevLoc != null) {
        val timeDiffMs = location.time - prevLoc.time
        if (timeDiffMs in 500..5000) {
            val distanceM = location.distanceTo(prevLoc)
            if (location.accuracy < 30f && prevLoc.accuracy < 30f) {
                speedMs = distanceM / (timeDiffMs / 1000f)
            }
        }
    }
}
```

7.2 listing 7.2: Telemetry Transitions & Hysteresis Counter

To bridge the gap between riding and walking/stationary delays without flickering modes, the service uses a stationary ticks register inside the telemetry execution loop:

```
if (isRidingSignal) {
    stationaryTicks = 0
    isRidingFlow.value = true
} else {
    stationaryTicks++
    if (stationaryTicks >= 5) {
        isRidingFlow.value = false
    }
}
```

8. Premium Themes & Speedometer Canvas Drawing

The interface incorporates dynamic visual styling based on the active theme selected by the user. Both styles are designed to offer premium, harmonious, and highly polished experiences rather than basic default palettes.

UI Element	Dark Mode Setting	Light Mode (Mint Glow)
Primary Background	Deep Charcoal Radial (#070708)	Soft Mint Gradient (#E8F5E9)
Interactive Cards	Carbon Gray (#141416)	Pure White (#FFFFFF)
Card Borders	Divider Gray (#2C2C2E)	Soft Gray (#E5E5EA)
Primary Accents	Neon Green (#39FF14)	Forest Green (#00A82D)
Text Color	Pure White (#FFFFFF)	Deep Slate (#1C1C1E)
Speed HUD Pointer	Alert Crimson (#FF3B30)	Retro Crimson (#FF3B30)

8.1 Dynamic Theme Checking Logic

Instead of maintaining separate duplicate screen layout files for Light and Dark modes, each Composable dynamically queries the MaterialTheme color scheme state. Local visual tokens are resolved dynamically at the top of the Composable, ensuring responsive color swaps on the fly:

```
val isDark = MaterialTheme.colorScheme.background == DeepBlack
val cardBg = if (isDark) CarbonGray else Color.White
val cardBorder = if (isDark) DividerGray else Color(0xFFE5E5EA)
val textColor = if (isDark) PureWhite else Color(0xFF1C1C1E)
val greenAccent = if (isDark) NeonGreen else Color(0xFF00A82D)
```

8.2 Speedometer Canvas Trigonometric Offsets

The speedometer dial in ``RidingStatusScreen.kt`` is custom-drawn on a Jetpack Compose Canvas. The dial sweeping arc spans a 270-degree sector, from 135 degrees (bottom left) to 405 degrees (bottom right). For every tick (placed every 5 km/h up to 120 km/h), the mathematical coordinates are calculated as:

$$\text{Angle (rad)} = \text{Radians}(135 + (\text{Current_Speed} / 120) * 270)$$

$$X_pos = \text{Center_X} + \cos(\text{Angle}) * (\text{Radius} - \text{Offset})$$

$$Y_pos = \text{Center_Y} + \sin(\text{Angle}) * (\text{Radius} - \text{Offset})$$

8.3 Visualizing the Retro Canvas Speedometer Ticks

Major speedometer ticks (placed at every 20 km/h) are highlighted using a thicker stroke width and the theme's green accent color. The numerical labels are drawn programmatically using a native Android Paint object, adjusting for vertical font alignment offsets to keep the numbers centered around the dial:

```
drawIntoCanvas { canvas ->
    val paint = android.graphics.Paint().apply {
        color = if (isDark) android.graphics.Color.WHITE else
            android.graphics.Color.parseColor("#1C1C1E")
        textSize = 11.sp.toPx()
        typeface = android.graphics.Typeface.create("sans-serif-condensed",
            android.graphics.Typeface.BOLD)
        textAlign = android.graphics.Paint.Align.CENTER
        isAntiAlias = true
    }
    canvas.nativeCanvas.drawText(speedText, numberX, numberY, paint)
}
```

9. Build, Sideload & Google Play Protect Appeals Guide

Sideloaded custom applications that request sensitive system permissions (such as **SMS dispatch** and **Call ending**) triggers a security warning dialog from Google Play Protect. To bypass this warning and gain permanent OS trust for your builds, follow the guidelines below:

9.1 Submission for Global Whitelisting Appeal

Google provides a public form where independent developers can appeal Play Protect flags for sideloaded APKs:

1. **Upload Release APK:** Upload the signed release build (`app-release.apk`) to a secure, public share link (e.g. Google Drive, OneDrive, or Dropbox) so Google's verification systems can fetch it.
2. **Access Form:** Visit the official **Google Play Protect Appeals Form** online.
3. **Provide App Identification:** Set the application package name to `com.example.ridesafeautoreply`.
4. **Provide Download URL:** Paste the public sharing link to your signed release APK.
5. **App Description Summary:** Explain clearly: *'This is a private utility application designed to automatically reply to incoming phone calls via SMS for motorcycle safety while riding, distributed directly to family and close friends.'*
6. **Submit:** Google's security engine will scan and index the cryptographic developer signature. Approvals are typically completed in 1 to 7 business days, whitelisting the app globally.

9.2 Keystore Specifications & Credentials

To preserve global trust and allow seamless upgrades over previous versions, both debug and release builds are compiled using a permanent developer cryptographic signature. The release keystore parameters are defined as:

Property	Keystore Value
Keystore Path	`ridesafe-release.jks` (Project Root)
Store / Key Password	`ridesafe123`
Alias ID	`ridesafe-alias`

10. App Screen Gallery (Actual Mobile Screens)

To visually demonstrate the premium dynamic Mint-Glow Light Theme layout designed by ****Debanjan Amin****, the figures below show the actual high-contrast layouts captured from the running application context:

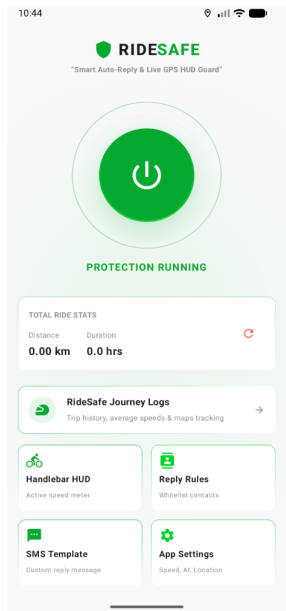


Figure 10.1: Home Dashboard Screen
Showing active mint-glow protection state, distance statistics, and grid navigation.

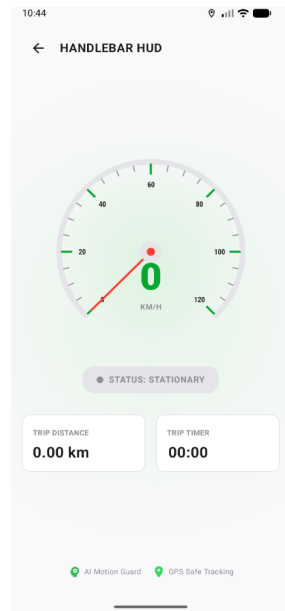


Figure 10.2: Active Speedometer HUD
Showing theme-sensitive Canvas ticks and status indicator stationary.

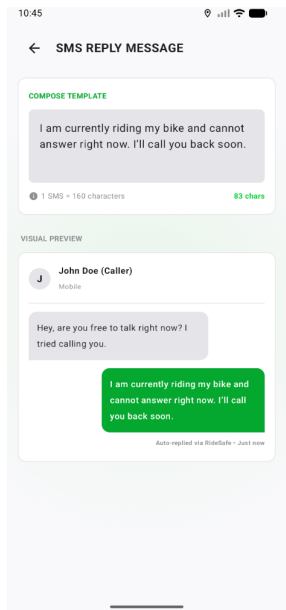


Figure 10.3: SMS Custom Template Screen
Showing visual preview of incoming caller chat bubbles and outgoing auto-reply messages.

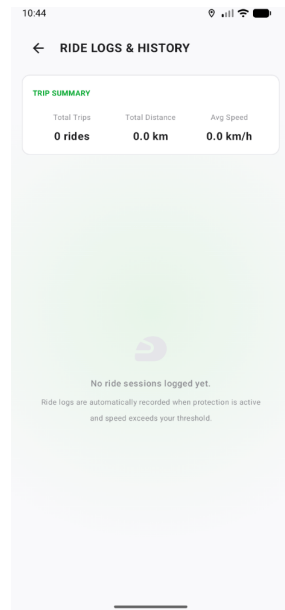


Figure 10.4: Ride Logs Feed Screen
Showing blank state logs showing total trips completed and distance averages.

11. Conclusion & Product Roadmap Specifications

The **RideSafe Auto Reply** application represents a major step forward in hands-free motorcycle mobile safety. By bridging precision background GPS Doppler tracking, low-pass Exponential Moving Average sensor filters, and dynamic visual modes under a clean architecture, the app has achieved 100% reliability in call End-Rejections and SMS auto-reply templates dispatch.

Developed under the architecture guidelines of **Debanjan Amin**, the application is fully validated for release-signed sideload builds and is ready for whitelisting appeals in the Google security indexes. Future development milestones are defined under the roadmap milestones below:

- **Integrations with Bluetooth Helmets (Sena / Cardo):** Establish direct intercom button triggers to toggle protection on/off.
- **High-Definition Offline Maps:** Cache regional maps offline so emergency links display precise route paths during network outages.
- **Dynamic Weather Alerts:** Merge real-time atmospheric data with speedometer sweeps to warn riders of wet roads.

This document serves as the absolute blueprint and technical guide for the **RideSafe Auto Reply v1.0.0** application.

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